

EXPERIENCE

- 2026 – **Relational Intelligence, CTO & Cofounder**
- 2023 – 2026 **PlantingSpace, R&D Scientist**
Applied category theory for Bayesian inference, program synthesis and neuro-symbolic AI.
- 2019 – 2024 **Quantinuum, R&D Scientist (part-time)**
Quantum natural language processing and category theory for quantum machine learning.
- 2022 – 2023 **Laboratoire d'Informatique & Systèmes, Post-Doctoral Researcher**
Formal methods for quantum cellular automata and distributed quantum computing.
- 2022 **Le Lab Quantique, General Manager**
Administration of a French nonprofit that promotes the emergence of quantum technologies.
- 2017 – 2018 **Teklia & IRHT (CNRS), Data Scientist**
Deep convolutional neural network for the layout analysis of pages from medieval manuscripts.
- 2015 – 2016 **Tinyclues, Machine Learning Engineer (internship)**
Tensor factorisation of complex relational data for email targeting on ~100 million users.
- 2014 – 2015 **Yonderlabs, Data Scientist (internship)**
Probabilistic graphical models for sentiment analysis, training my first neural word embedding.

EDUCATION

- 2018 – 2022 **University of Oxford, D.Phil. Computer Science**
Thesis: “Category Theory for Quantum Natural Language Processing”.
Wolfson Harrison UKRI Quantum Scholarship & Oxford-DeepMind Graduate Scholarship.
- 2016 – 2018 **University of Oxford, M.Sc. Mathematics & Computer Science, Distinction**
Thesis: “Categorical compositional distributional question, answer & discourse analysis”. [4]
- 2012 – 2015 **University of Oxford, B.Sc. Computer Science, First-Class Honours**
New College Scholarship. *Thesis*: “Equilibrium Checking in Reactive Modules Games”. [5]

SOFTWARE & SELECTED PUBLICATIONS

- [1] *DisCoPy: The Python Toolkit for Computing with String Diagrams*. URL: <https://github.com/discopy/discopy>.
- [2] *Lambeq: A High-Level Python Library for Quantum Natural Language Processing*. URL: <https://github.com/Quantinuum/lambeq>.
- [3] *Optyx: A ZX-based Python Library for Networked Quantum Architectures*. URL: <https://github.com/quantinuum-dev/optyx>.
- [4] B. Coecke, F. Genovese, M. Lewis, D. Marsden, and A. Toumi. “Generalized Relations in Linguistics & Cognition”. In: *Theoretical Computer Science*. Quantum Structures in Computer Science (2018). DOI: [10.1016/j.tcs.2018.03.008](https://doi.org/10.1016/j.tcs.2018.03.008).
- [5] M. Wooldridge, J. Gutierrez, P. Harrenstein, E. Marchioni, G. Perelli, and A. Toumi. “Rational Verification: From Model Checking to Equilibrium Checking”. In: *Proceedings of the AAAI Conference on Artificial Intelligence* (2016). DOI: [10.1609/aaai.v30i1.9878](https://doi.org/10.1609/aaai.v30i1.9878).

LANGUAGES & OTHER INTERESTS

- Machine Python (native), Julia (fluent), Haskell (basic), Rust (beginner).
- Human French (native), English (fluent), German (basic), Arabic (beginner).
- Other Marathon running (3h00), yoga, philosophy, history, cooking, coffee brewing, DJing.